

EXP NO: 02

HALF ADDER AND FULL ADDER

DATE:

27/07/2026

Aim:

To realize half / full adder

- i) Using X-OR and basic gates.
- ii) Using only NAND gates.

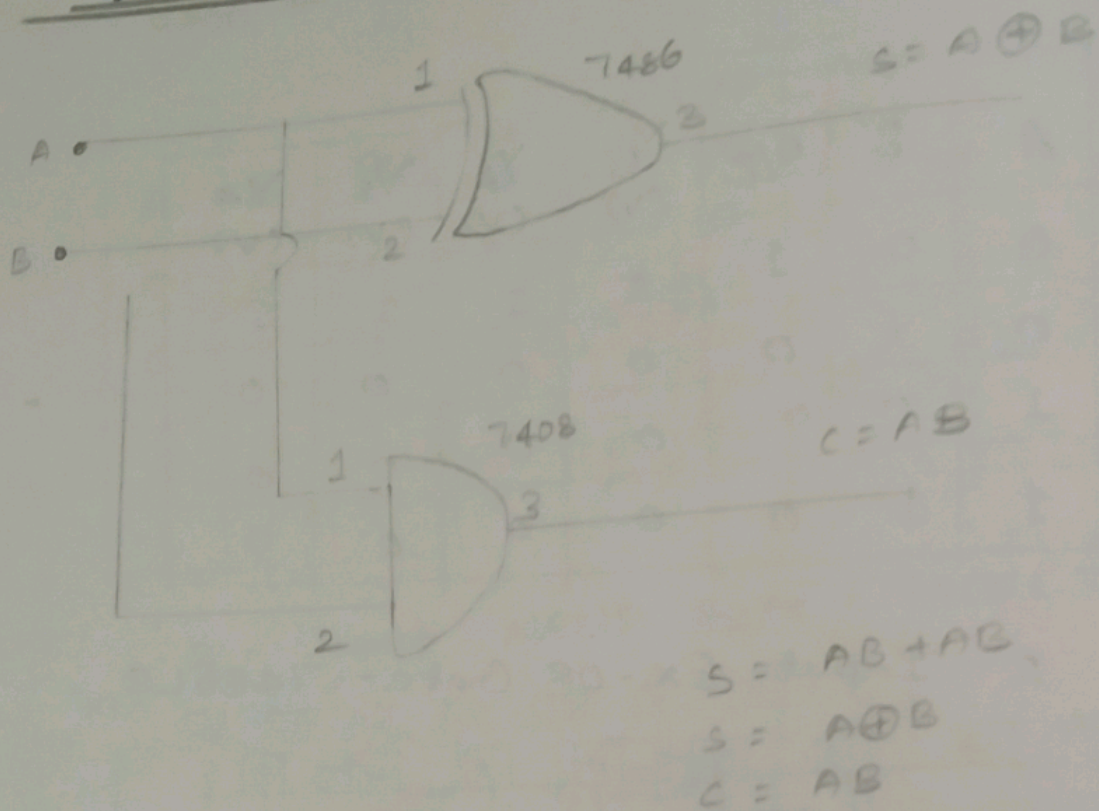
Apparatus Required

Digital Trainer Kit, IC 7486,
IC 7432, IC 7408, IC 7400 and connecting
wires.

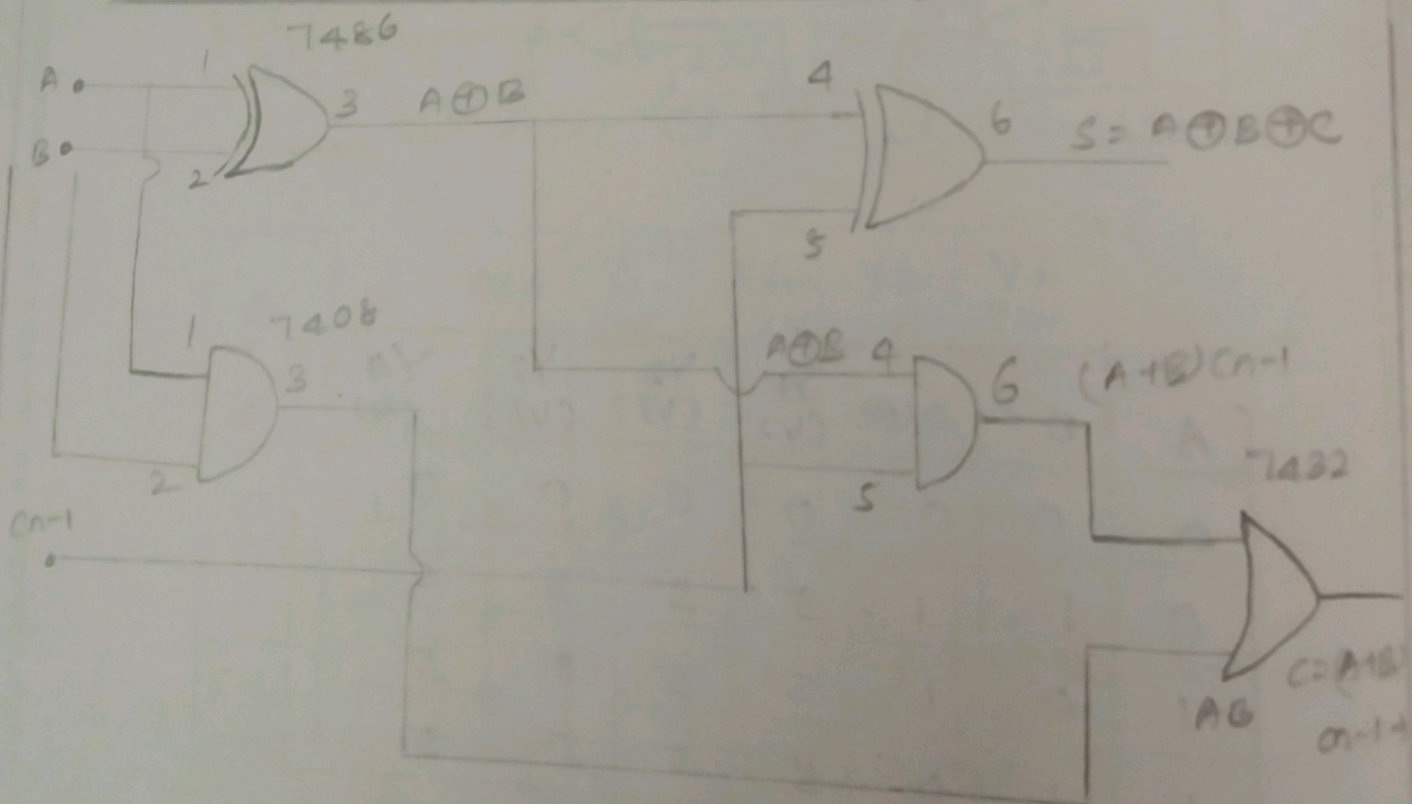
Procedure

1. Verify the gates
2. Make the connections as per the circuit diagram.
3. Switch on Vcc and apply various combinations of input according to the truth table.
4. Note down the output readings for half and full adder sum, difference and the carry / borrow bit for different combinations of inputs.

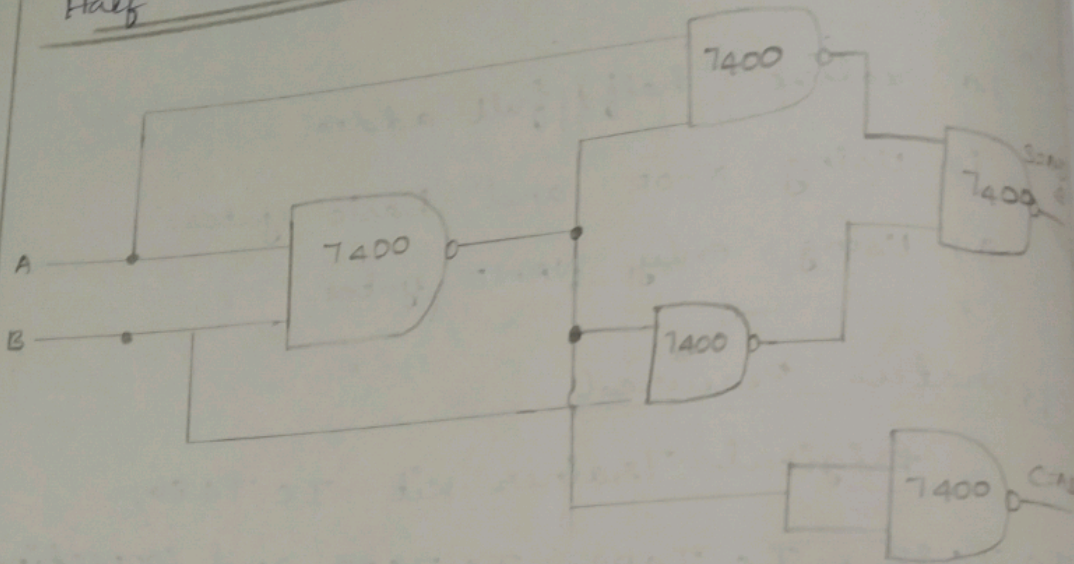
Half Adder using basic gates



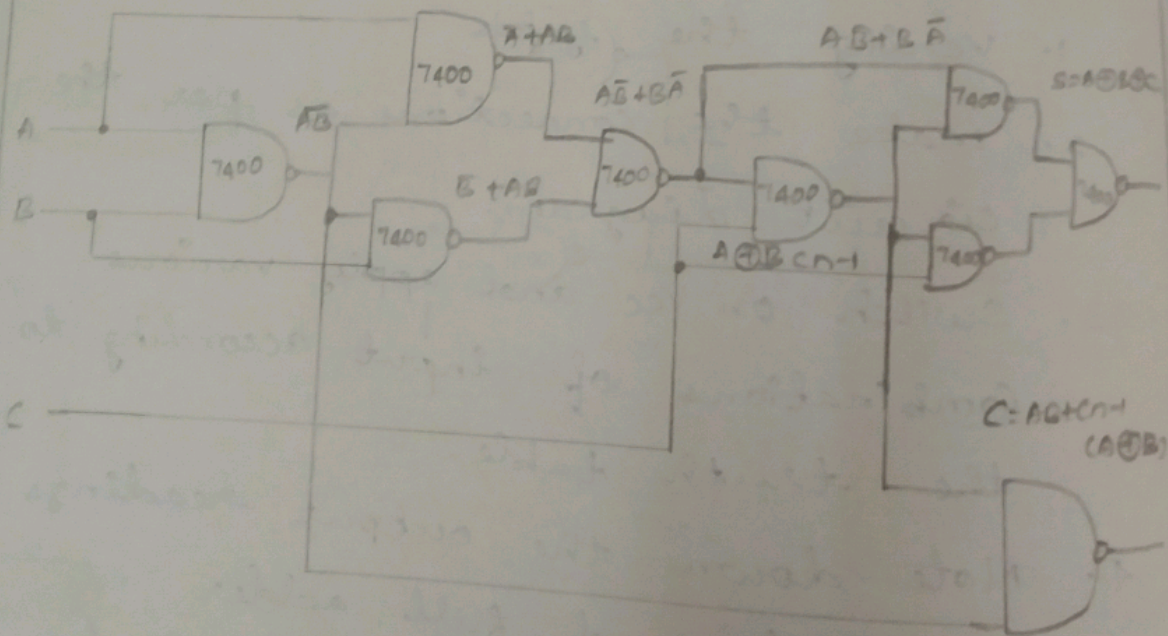
Full adder Using basic gates



Half Adder using NAND gate only



Full Adder using NAND gates only :-



Half Adder

Half

A	B
0	0
0	1
1	0
1	1

Full Adder

Fu

A	B	C
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

Half Adder

Half Adder					
A	B	S	C	SCV	CCV
0	0	0	0	0	0
0	1	1	0	5	0
1	0	1	0	5	0
1	1	0	1	0	5

Full Adder

Full Adder						
A	B	C _{n-1}	S	C	SCV	CCV
0	0	0	0	0	0	0
0	0	1	1	0	5	0
0	1	0	1	0	5	0
0	1	1	0	1	0	5
1	0	0	1	0	5	0
1	0	1	0	1	0	5
1	1	0	0	1	0	5
1	1	1	1	1	5	5